

HW4

Execution description

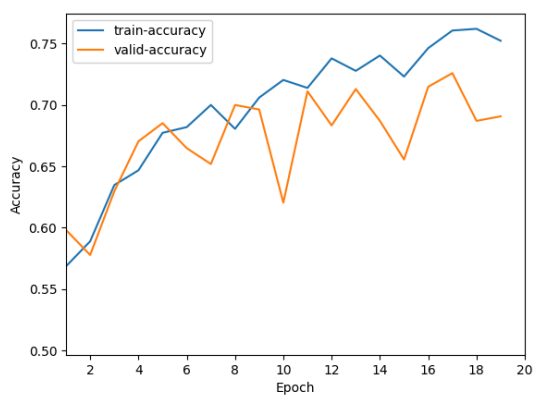
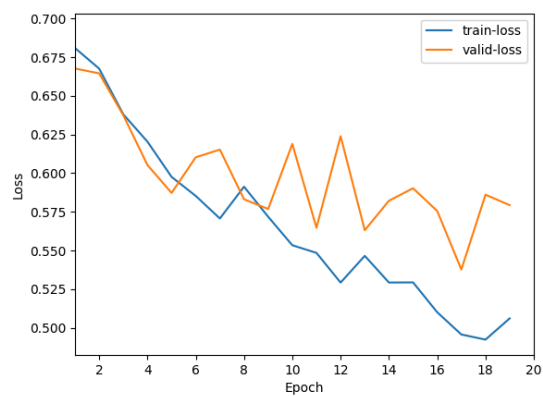
```
./train_grid.sh
```

Experimental results

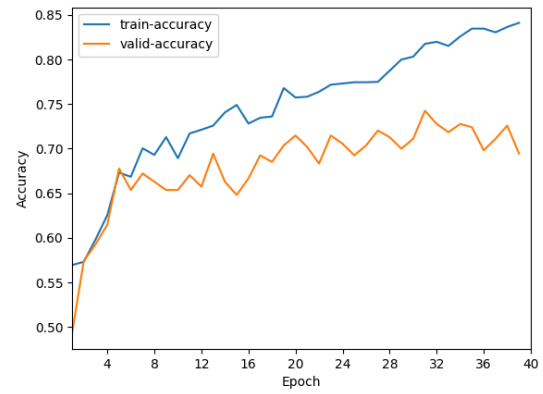
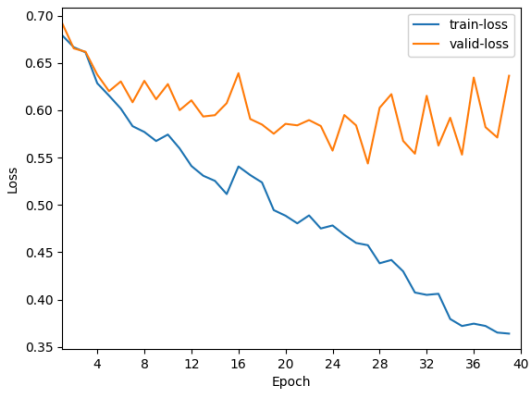
Epochs 20 / 40 / 80

Loss_curve / Training_accuracy

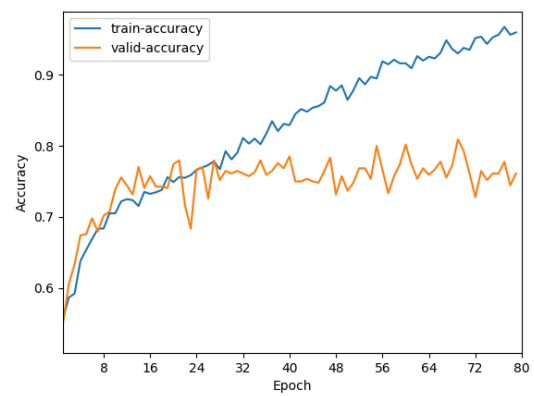
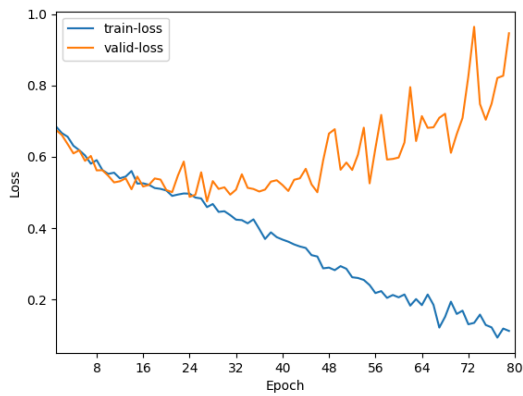
Epochs: 20



Epochs: 40



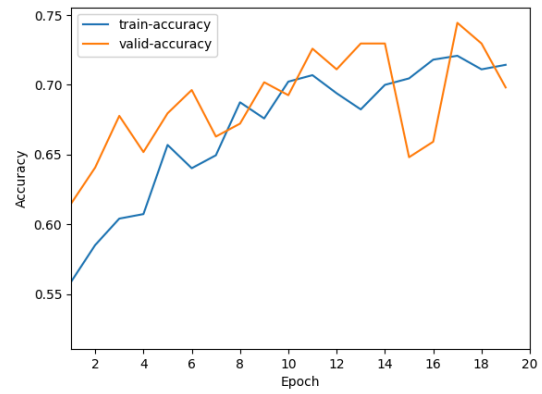
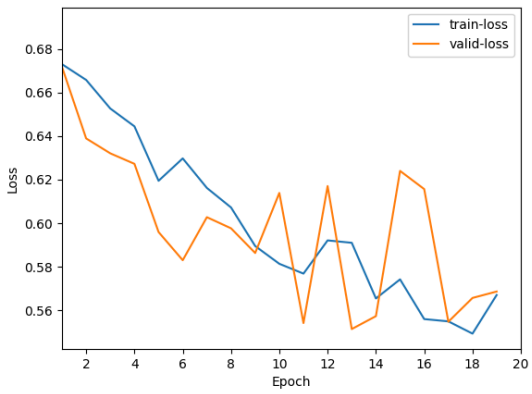
Epochs: 80



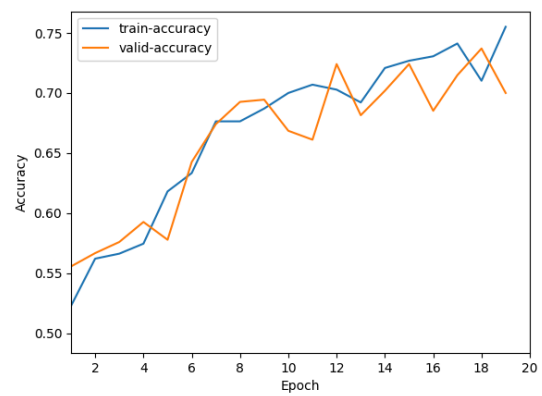
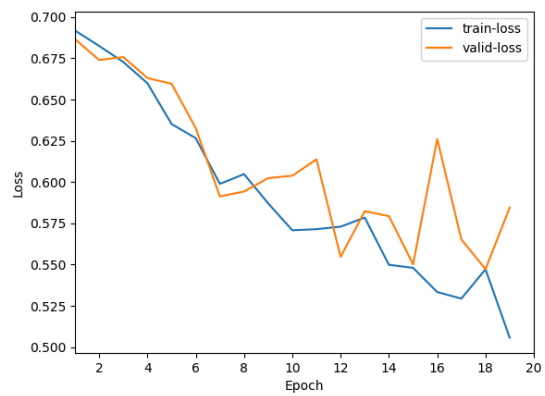
Batch size 8 / 16 / 32

Loss_curve / Training_accuracy

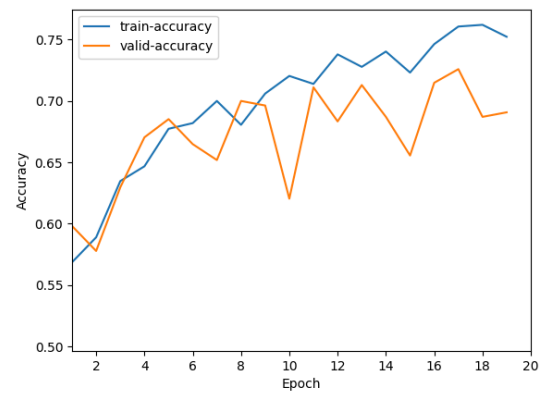
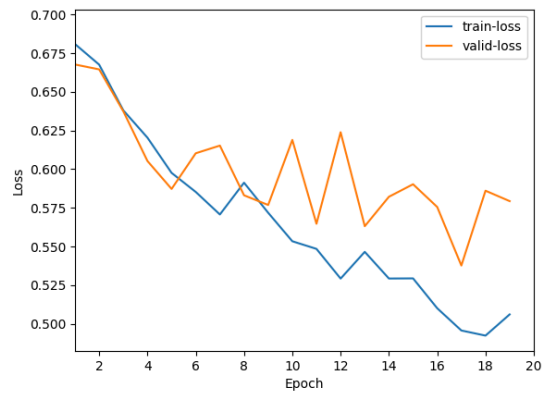
Batch size: 8



Batch size: 16



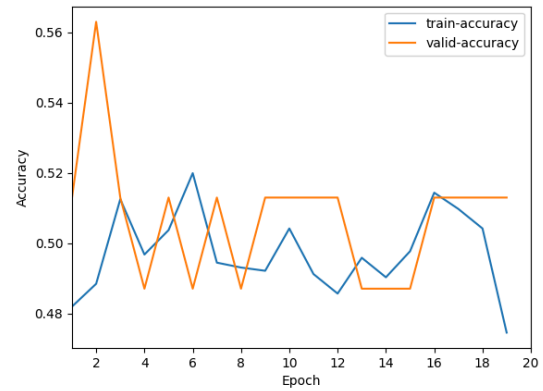
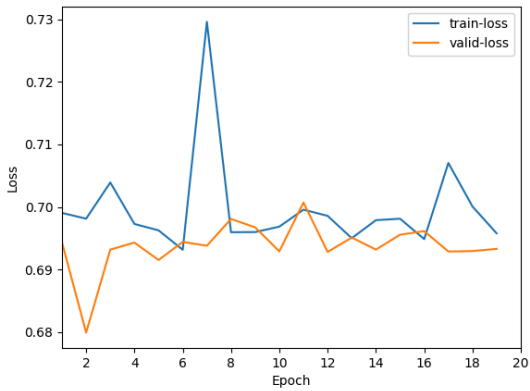
Batch size: 32



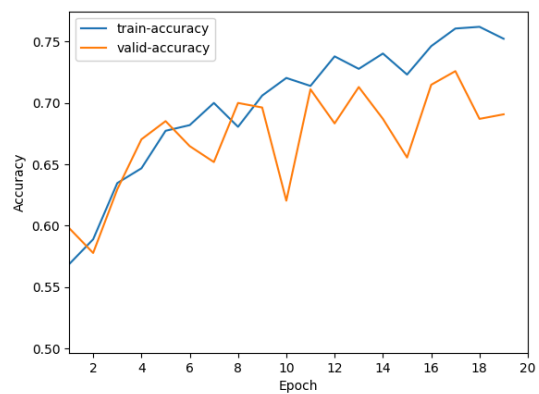
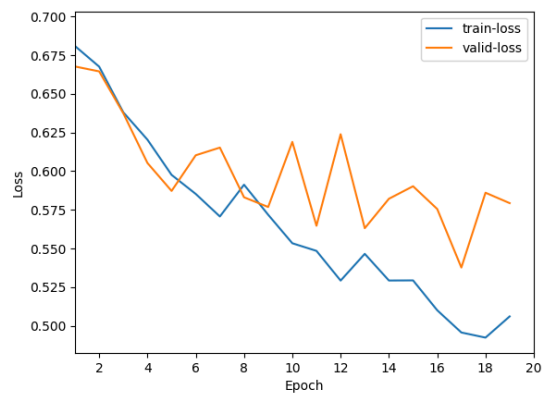
Learning Rate 0.1 0.01 0.001

Loss_curve / Training_accuracy

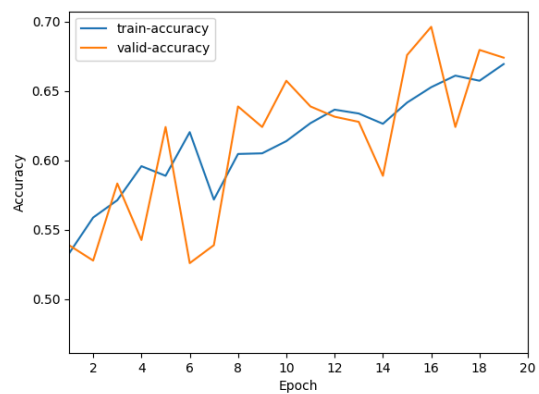
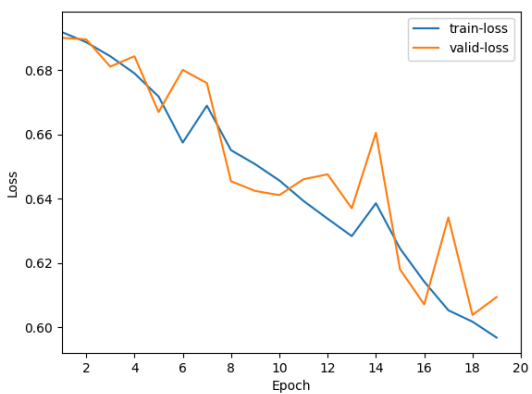
Learning Rate: 0.1



Learning Rate: 0.01

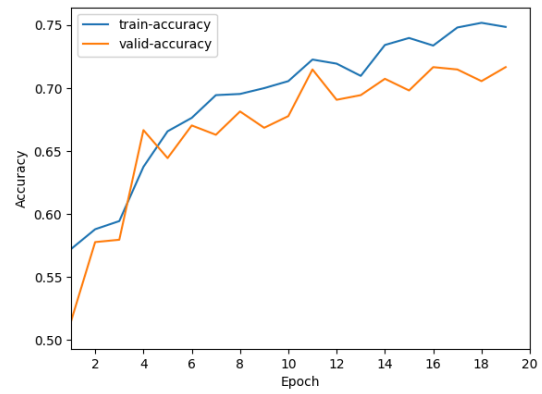
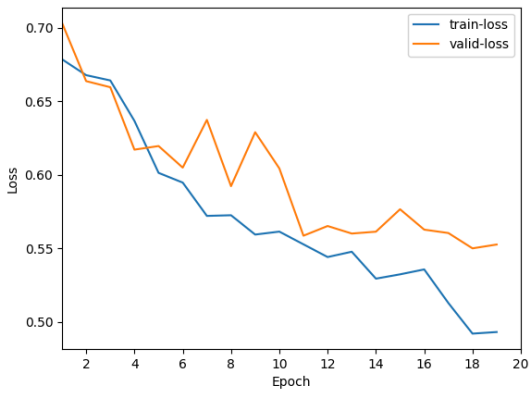


Learning Rate: 0.001

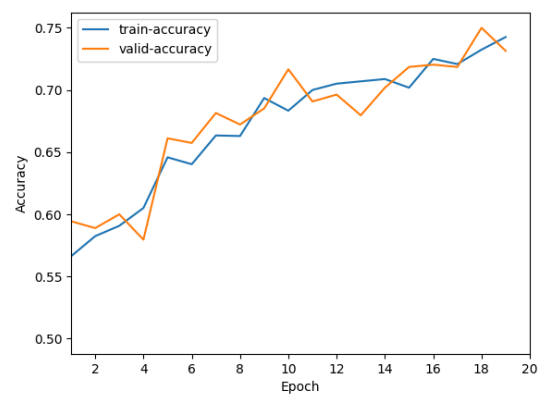
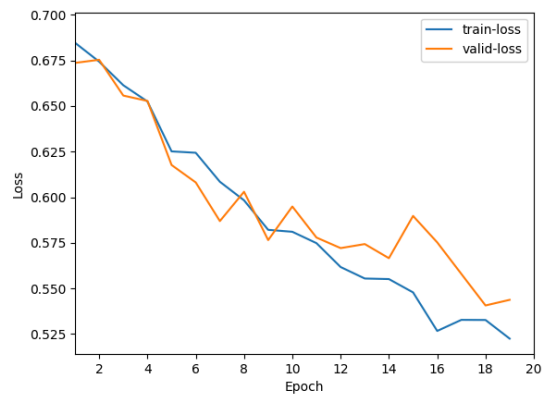


Loss CrossEntropyLoss, BCEWithLogitsLoss, FocalLoss

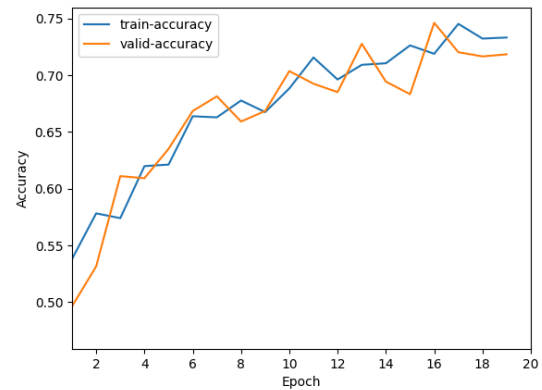
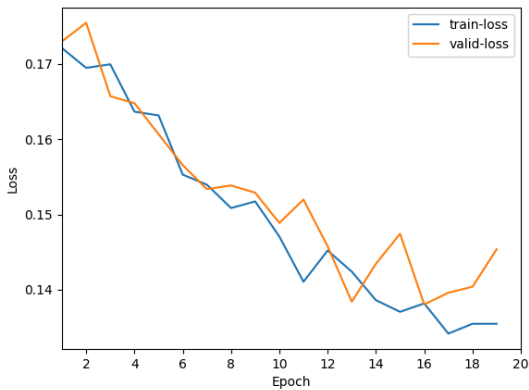
CrossEntropyLoss



BCEWithLogitsLoss



FocalLoss



Result

```
{ "weight_path": "weight_e20_bs32_lr0.01.pth", "accuracy": 74.0 }
{ "weight_path": "weight_e40_bs32_lr0.01.pth", "accuracy": 78.0 }
{ "weight_path": "weight_e80_bs32_lr0.01.pth", "accuracy": 73.3333 }
{ "weight_path": "weight_e20_bs8_lr0.01.pth", "accuracy": 66.3333 }
{ "weight_path": "weight_e20_bs16_lr0.01.pth", "accuracy": 72.0 }
{ "weight_path": "weight_e20_bs32_lr0.1.pth", "accuracy": 55.0 }
{ "weight_path": "weight_e20_bs32_lr0.001.pth", "accuracy": 67.3333 }
{ "weight_path": "weight_loss_ce.pth", "accuracy": 71.0 }
{ "weight_path": "weight_loss_bce.pth", "accuracy": 69.0 }
{ "weight_path": "weight_loss_focal.pth", "accuracy": 70.6667 }
```

Conclusion

調整項目	調整效果與觀察	優點	缺點
Epoch 數量	增加 epoch 可讓模型學習更充分，Loss 持續下降，Accuracy 提升。但 epoch 過多可能導致 overfitting。	模型學得更完整，準確率提升	訓練時間拉長，過多會過擬合

Batch Size	太小會導致 Loss 曲線波動較大、訓練時間變長，但泛化能力好；太大雖然訓練穩定、速度快，但可能收斂太快導致測試效果差。	小：泛化能力好； 大：訓練快、記憶體利用佳	小：不穩定、慢； 大：可能過早收斂、泛化較差
Learning Rate	過高會讓 Loss 曲線劇烈震盪、不收斂；過低則收斂過慢，難以有效訓練。適當值（如 0.01）能穩定提升準確率。	適當值可快速且穩定學習	過高學不到東西；過低訓練太久，可能卡在局部最小值
Loss Function	CrossEntropy 適用於分類，穩定好用；BCE 適合二分類任務；FocalLoss 處理不平衡數據效果好，但收斂慢。	CrossEntropy：穩定好用；FocalLoss：處理難分類樣本佳	FocalLoss 收斂慢；BCE 需注意搭配輸出層設計

Dicussion

這次跑好久... 如果沒寫一個 bash 檔案真的要一直重複跑很躁, 最後爆改了一堆原本的 example code 才方便了許多